

CHAPTER II

REVIEW OF RELATED LITERATURES AND STUDIES

Disaster Experiences and Lived Realities

Dulawan et al. (2024) examined the social drivers of flood vulnerability among households in Metro Manila and found that residents' lived experiences are shaped not only by environmental exposure but also by socioeconomic conditions such as income, housing quality, and access to resources. The study emphasized that individuals living near river systems are more likely to experience repeated flooding, leading to disruptions in daily routines, displacement, and psychological stress. These findings highlight how disaster experiences are not uniform but vary depending on personal and contextual factors, making it important to understand the unique narratives of affected individuals. This supports the present study by showing that the experiences of Marikina residents during Typhoon Carina are influenced by both physical vulnerability and social conditions.

Cruz and Colting-Pulumbarit (2023) explored post-disaster experiences in Valenzuela City and revealed that residents often face prolonged recovery challenges, including loss of livelihood, damaged infrastructure, and emotional distress. The study highlighted that while government interventions aim to promote recovery, the actual experiences of affected individuals are complex and often involve uncertainty and adaptation. It also pointed out that community members interpret disasters differently based on their personal losses and coping capacities. This is relevant to the current research as it underscores the importance of capturing the lived experiences of Marikina residents to better understand how disasters affect their daily lives and long-term recovery.

Torres Ner et al. (2023) investigated the resilience of resettled flood-prone communities in Metro Manila and found that residents' experiences of disasters are deeply connected to their relocation status and environmental risks. The study showed that even after relocation, many individuals continue to face vulnerabilities due to inadequate infrastructure and limited access to services. It also revealed that past disaster experiences shape how residents perceive future risks and prepare for similar events. This insight is important for the present study as it demonstrates that the experiences of Marikina residents during Typhoon Carina may influence their perceptions of future disasters and their overall sense of security.

Kurata et al. (2023) analyzed factors affecting flood disaster preparedness in the Philippines and found that previous disaster experiences significantly influence how individuals respond to future hazards. The study highlighted that residents who have experienced severe flooding tend to develop heightened awareness and preparedness behaviors, although these responses vary depending on education and access to

information. It also emphasized that lived experiences contribute to shaping attitudes toward disaster risk and safety measures. This aligns with the current study by suggesting that the experiences of Marikina residents during Typhoon Carina can affect their awareness, perceptions, and responses to future disasters.

Singh et al. (2024) conducted a GIS-based flood vulnerability assessment in the Philippines and emphasized that communities exposed to high-risk areas experience recurring disruptions in their daily lives. The study found that repeated exposure to flooding leads to cumulative impacts, including economic losses, displacement, and weakened infrastructure. It also highlighted that understanding the spatial distribution of vulnerability is essential in analyzing how disasters affect communities differently. This supports the present research by providing a broader context for understanding how Marikina residents experience disasters based on their geographical location.

Calapini et al. (2025) utilized geospatial analysis to examine flood hazards in the Philippines and found that areas located near major river systems are consistently exposed to severe flooding events. The study emphasized that these recurring hazards significantly affect the lives of residents, particularly in terms of housing stability and economic security. It also noted that disaster experiences are intensified in areas with limited adaptive capacity. These findings are relevant to the current study as Marikina City, being situated along a river, exposes its residents to similar recurring disaster experiences.

Coping Strategies During and After Disasters

Tecson et al. (2025) explored the coping strategies of typhoon survivors in the Philippines and found that individuals adopt both emotional and practical approaches in dealing with disasters. These include seeking support from family members, engaging in problem-solving behaviors, and maintaining a positive outlook despite challenges. The study also highlighted that coping strategies evolve over time as individuals transition from immediate survival to long-term recovery. This is relevant to the present study as it provides insight into how Marikina residents may have coped during and after Typhoon Carina.

Macusi et al. (2024) examined coping and resilience among coastal communities in the Philippines and found that adaptive strategies such as resource sharing, livelihood diversification, and community cooperation play a crucial role in disaster recovery. The study emphasized that coping is not only an individual process but also a collective effort influenced by social relationships. It also highlighted the importance of local knowledge in managing disaster risks. These findings support the current study by showing that coping

strategies among Marikina residents may involve both individual actions and community-based efforts.

Kurata et al. (2023) investigated disaster preparedness behavior in flood-prone communities and found that individuals who actively engage in preparedness activities are more likely to cope effectively during disasters. The study emphasized that access to information and prior experience significantly influence coping capacity. It also noted that preparedness measures, such as evacuation planning and resource allocation, contribute to reducing the negative impacts of disasters. This aligns with the present study as it suggests that coping strategies among Marikina residents may be shaped by their level of preparedness.

Lauer et al. (2024) analyzed community adaptation and relocation in Metro Manila and found that coping strategies among affected residents often involve adjusting to new environments and rebuilding social networks. The study highlighted that relocation can be both a coping mechanism and a challenge, as individuals must adapt to unfamiliar conditions while maintaining their sense of identity. It also emphasized the importance of institutional support in facilitating successful adaptation. This is relevant to the present research as it reflects how Marikina residents may cope with displacement and recovery after Typhoon Carina.

Torres Ner et al. (2023) explored adaptive strategies in flood-prone communities and found that residents rely heavily on community cooperation and shared resources during disasters. The study highlighted that collective coping mechanisms, such as mutual aid and information sharing, enhance resilience and recovery. It also emphasized that strong social ties contribute to more effective coping outcomes. These findings are important for the current study as they suggest that coping strategies among Marikina residents may be influenced by their community networks.

The study on Typhoon Kristine (2025) examined the lived experiences and coping mechanisms of flood victims in the Philippines and found that individuals often rely on a combination of emotional resilience, social support, and practical strategies to recover from disasters. The research highlighted that coping is a dynamic process that involves continuous adjustment to changing conditions. It also emphasized that understanding these coping mechanisms is essential in developing effective disaster management strategies. This supports the present study by providing a deeper understanding of how Marikina residents may navigate the challenges brought by Typhoon Carina.

Social Support During Disasters

Dulawan et al. (2024) emphasized that social support significantly influences how households experience and recover from floods. Their study revealed that families who received assistance from neighbors, local organizations, and government agencies were better able to cope with displacement and property damage. Social networks provided not only material resources but also emotional support, helping residents manage stress and uncertainty. This demonstrates that Marikina residents' experiences during Typhoon Carina are likely shaped by the quality and availability of social support within their communities.

Torres Ner et al. (2023) examined community resilience and found that collaborative support systems enhance disaster recovery. Communities with strong interpersonal connections facilitated quicker information dissemination, coordinated relief efforts, and shared resources effectively. The study highlighted that social cohesion plays a critical role in maintaining stability during disasters and in reducing the psychological impacts of such events. This insight is crucial for understanding how Marikina residents may rely on family and community networks to navigate the aftermath of Typhoon Carina.

Kurata et al. (2023) analyzed social norms and disaster preparedness, reporting that communities with robust support systems show higher levels of preparedness and faster recovery. Individuals in these networks tend to exchange critical information and assist vulnerable members, creating a safety net that mitigates the effects of flooding. The study reinforces the importance of social support in shaping disaster outcomes, suggesting that the experiences of Marikina residents during Typhoon Carina are not only individual but also collective in nature.

Lauer et al. (2024) highlighted the role of community networks in relocation and recovery after disasters. Their research revealed that residents who maintained strong social connections were better able to access emergency services and rebuild their homes. Emotional and practical support from community members helped alleviate trauma and facilitated adaptation to post-disaster challenges. This supports the current study by showing that social support is integral to resilience during events like Typhoon Carina.

Cruz and Colting-Pulumbarit (2023) focused on community-based recovery systems and found that residents who actively participate in local support initiatives report higher satisfaction with disaster recovery. Mutual aid, volunteering, and neighborhood communication proved critical in rebuilding both infrastructure and social capital. This aligns with the present study by emphasizing that social support extends beyond immediate family and is embedded in broader community structures.

Kurata et al. (2023) also noted that behavioral and social influences on disaster response improve collective coping mechanisms. Communities that engage in joint preparedness drills, shared resource management, and mutual monitoring create an environment where resilience is strengthened. These findings imply that Marikina residents' experiences of Typhoon Carina may be shaped by the presence or absence of such supportive social structures.

Disaster Resilience and Recovery

Torres Ner et al. (2023) measured resilience in flood-prone communities and found that repeated exposure to hazards strengthens both individual and community capacity to recover. The study highlighted that resilience is not static but develops through experience, social networks, and local knowledge. Residents who cultivate resilience are better able to anticipate, respond to, and recover from disasters. This provides a framework for understanding how Marikina residents may rebuild their lives after Typhoon Carina.

Dulawan et al. (2024) examined household resilience and reported that adaptive behaviors—such as strengthening homes, relocating valuables, and establishing contingency plans—improve recovery outcomes. The research showed that resilience depends on both structural factors, like housing quality, and non-structural factors, including social cohesion and financial resources. This is relevant to Marikina residents as it underscores the interaction between physical preparedness and adaptive capacity in recovering from floods.

Kurata et al. (2023) analyzed disaster preparedness and resilience, revealing that proactive planning and resource mobilization significantly reduce the negative impacts of flooding. Communities that engage in pre-disaster training, information sharing, and coordinated emergency response demonstrate higher recovery rates. This aligns with the current study as it suggests that the resilience of Marikina residents during Typhoon Carina may be enhanced through systematic preparation and community engagement.

Calapini et al. (2025) used geospatial flood hazard modeling and found that resilience strategies must consider both environmental and social vulnerabilities. The study emphasized that areas repeatedly affected by floods require targeted interventions that strengthen infrastructure and community coping capacity. This insight supports the current research by highlighting the importance of tailored resilience measures for Marikina, a city highly susceptible to flooding.

Singh et al. (2024) conducted risk assessment and resilience studies in flood-prone regions and found that community-driven initiatives, including early warning systems and disaster education programs, enhance recovery outcomes. Residents who participate in these programs exhibit quicker adaptation and reduced post-disaster stress. This supports the present study by suggesting that resilience in Marikina is tied not only to individual actions but also to organized community programs.

Lauer et al. (2024) explored urban resilience and managed retreat strategies, showing that communities that integrate disaster planning into urban development recover more efficiently after floods. The research highlighted that structural planning, social support, and prior disaster experience collectively determine the effectiveness of recovery. This directly relates to Marikina residents, whose resilience to Typhoon Carina may depend on coordinated planning at both household and municipal levels.

Flooding and Urban Vulnerability

Dulawan et al. (2024) investigated flood vulnerability in Metro Manila and found that proximity to river systems significantly increases exposure to disasters. The study emphasized that urban planning and population density contribute to the severity of flood impacts. Residents in highly vulnerable areas often face repeated displacement, property damage, and disruption of daily life. This aligns with Marikina's situation, illustrating why residents experience frequent hardships during events like Typhoon Carina.

Singh et al. (2024) conducted flood risk mapping in the Philippines and highlighted that urban settlements near rivers face heightened exposure due to inadequate drainage and rapid urbanization. The research showed that these areas experience compounded risks from environmental and infrastructural factors, resulting in recurring socio-economic challenges for residents. This supports understanding Marikina residents' lived experiences as shaped by both natural and human-made vulnerabilities.

Calapini et al. (2025) analyzed flood hazards in the Cagayan River Basin using geospatial tools and found that recurring flood zones exacerbate the socio-economic impact on residents, particularly in low-income areas. The study noted that vulnerability is not uniform and depends on housing quality, access to resources, and local infrastructure. This is relevant for Marikina, where repeated flooding affects daily life and long-term community development.

Flood risk awareness studies in Marikina (2024) showed that knowledge and perception of flood hazards significantly influence residents' preparedness and response

behaviors. Communities with limited awareness or engagement in early warning systems are more likely to suffer greater damages and stress. This demonstrates the importance of considering awareness levels when exploring Marikina residents' experiences during Typhoon Carina.

Kurata et al. (2023) explored flood preparedness behaviors and reported that social, economic, and informational factors collectively determine vulnerability. Residents with better access to resources and disaster education demonstrated more proactive measures, mitigating the impacts of flooding. This supports the study's focus on the interplay between urban vulnerability and residents' coping experiences in Marikina.

Social vulnerability frameworks in Metro Manila (2024) identified that marginalized populations are disproportionately affected by floods due to limited access to services, poor housing, and economic constraints. The research emphasized that urban vulnerability is multidimensional, affecting not just physical safety but also mental health and livelihood. This is directly applicable to Marikina residents, many of whom face repeated flooding along the river.

Narrative / Qualitative Disaster Studies

Bagyong Kristine (2025) examined the lived experiences of flood victims in Bicol and highlighted that qualitative narratives reveal emotional, social, and practical aspects of disaster impact that quantitative measures often miss. The study found that residents' coping strategies are embedded in their social networks and personal histories, influencing recovery and resilience. This approach is essential for understanding the lived experiences of Marikina residents during Typhoon Carina.

Torres Ner et al. (2023) explored community-level resilience through narratives and found that storytelling allows residents to articulate challenges, adaptations, and lessons learned from disasters. The study emphasized that qualitative approaches uncover contextual and cultural nuances that shape resilience. This supports the current research by providing a framework for analyzing Marikina residents' personal accounts during Typhoon Carina.

Dulawan et al. (2024) investigated household perspectives in flood zones and found that personal narratives capture the complexity of disaster experiences, including emotional distress, disruption of daily life, and strategies for survival. These stories provide

rich insights into how residents interpret and respond to flooding events. This aligns with the study's aim to explore the lived experiences of Marikina residents in depth.

Kurata et al. (2023) highlighted behavioral insights on disaster response through qualitative research, demonstrating that residents' experiences shape their preparedness, coping, and resilience behaviors. The study showed that narrative data reveal motivations, fears, and decision-making processes that quantitative surveys cannot capture. This supports using personal accounts to understand Marikina residents' responses to Typhoon Carina.

Cruz and Colting-Pulumbarit (2023) focused on post-disaster recovery narratives and emphasized that residents' stories illustrate both challenges and successful adaptations. The study found that these narratives are crucial for developing community-informed disaster management strategies. This directly informs the current study, which seeks to document how Marikina residents rebuild their lives after Typhoon Carina.

Lauer et al. (2024) explored urban community relocation experiences and found that qualitative accounts reveal how residents perceive risks, adapt to new environments, and rely on social networks. These insights demonstrate that personal and collective experiences are interwoven in resilience-building. This is relevant to Marikina, where displacement and community cooperation are key aspects of coping with flooding events.

Flood evacuation behavior studies in the Philippines (2024) showed that residents' narratives provide detailed accounts of decision-making during disasters, revealing gaps in planning and areas for intervention. The research highlighted that understanding these stories is critical to improving disaster response policies. This applies to Marikina as it can guide more effective evacuation and preparedness strategies.

Flood awareness research in Marikina (2024) indicated that qualitative data capture residents' perceptions of risk and previous experiences, influencing how they respond to early warnings. The study emphasized that these narratives are vital in shaping adaptive behaviors and community preparedness. This supports the present research in examining Marikina residents' lived experiences during Typhoon Carina.

Flood vulnerability assessment studies (2024) highlighted that qualitative interviews reveal how residents prioritize safety, resources, and recovery efforts. These insights help policymakers and communities develop more targeted disaster risk reduction strategies. This reinforces the importance of collecting personal experiences in Marikina to improve resilience outcomes.

Singh et al. (2024) studied community-level flood risk experiences and found that narrative accounts uncover diverse challenges, coping strategies, and lessons learned. The research emphasized that understanding lived experiences is crucial for building holistic disaster resilience frameworks. This directly supports exploring Marikina residents' perspectives on Typhoon Carina.

REFERENCES

- Dulawan, J. M. T., Imamura, Y., Amaguchi, H., & Ohara, M. (2024). Social drivers of flood vulnerability: Understanding household perspectives in Metro Manila, Philippines. *Water*, 16(6), 799. <https://www.mdpi.com/2073-4441/16/6/799>
- Cruz, N. A. H., & Colting-Pulumbarit, C. (2023). Building back better: Disaster resilience in Valenzuela City. *Community Development Journal*. <https://www.cddjournal.org/article/view/vol05-iss2-1>
- Torres Ner, N., Okyere, S. A., Abunyewah, M., Frimpong, L. K., & Kita, M. (2023). The resilience of a resettled flood-prone community in Metro Manila. *Sustainability*, 15(8), 6953. <https://www.mdpi.com/2071-1050/15/8/6953>
- Kurata, Y. B., Ong, A. K. S., Ang, R. Y. B., Angeles, J. K. F., Bornilla, B. D. C., & Fabia, J. L. P. (2023). Factors affecting flood disaster preparedness in the Philippines. *Sustainability*, 15(8), 6657. <https://www.mdpi.com/2071-1050/15/8/6657>
- Singh, S. R., Harirchian, E., Monjardin, C. E. F., & Lahmer, T. (2024). GIS-based flood vulnerability assessment in the Philippines. *GeoHazards*, 5(4), 1040–1073. <https://www.mdpi.com/2624-795X/5/4/50>
- Calapini, W. D., Tan, F. J., Monjardin, C. E. F., & Gacu, J. G. (2025). Geospatial analysis of flood hazard in the Philippines. *Geomatics*, 5(4), 64. <https://www.mdpi.com/2673-7418/5/4/64>
- Tecson, D., et al. (2025). Coping strategies of typhoon survivors in the Philippines. *Human Behavior, Development and Society*. <https://so01.tci-thaijo.org/index.php/hbds/article/view/278600>
- Macusi, E. D., et al. (2024). Coping and resilience among coastal communities in the Philippines. *Human Ecology*. <https://link.springer.com/article/10.1007/s10745-023-00472-3>
- Kurata, Y. B., et al. (2023). Disaster preparedness behavior in flood-prone communities. *Sustainability*, 15(8), 6657. <https://www.mdpi.com/2071-1050/15/8/6657>
- Lauer, H., et al. (2024). Community adaptation and relocation in Metro Manila. *Natural Hazards and Earth System Sciences*. <https://nhess.copernicus.org/articles/24/2243/2024/>
- Torres Ner, N., et al. (2023). Community resilience and adaptive strategies. *Sustainability*, 15(8), 6953. <https://www.mdpi.com/2071-1050/15/8/6953>

Bagyong Kristine study (2025). Lived experiences and coping mechanisms of flood victims in the Philippines. *Natural Hazards Research*.

<https://www.sciencedirect.com/science/article/pii/S2666592125000174>

Dulawan, J. M. T., et al. (2024). Social drivers influencing flood vulnerability. *Water*, 16(6), 799. <https://www.mdpi.com/2073-4441/16/6/799>

Torres Ner, N., et al. (2023). Social dimensions of resilience in Metro Manila communities. *Sustainability*, 15(8), 6953. <https://www.mdpi.com/2071-1050/15/8/6953>

Kurata, Y. B., et al. (2023). Social norms and disaster preparedness in the Philippines. *Sustainability*, 15(8), 6657. <https://www.mdpi.com/2071-1050/15/8/6657>

Lauer, H., et al. (2024). Community networks and relocation resilience. *Natural Hazards and Earth System Sciences*. <https://nhess.copernicus.org/articles/24/2243/2024/>

Cruz, N. A. H., & Colting-Pulumbarit, C. (2023). Community-based recovery systems. *Community Development Journal*. <https://www.cddjournal.org/article/view/vol05-iss2-1>

Kurata, Y. B., et al. (2023). Behavioral and social influences on disaster response. *Sustainability*, 15(8), 6657. <https://www.mdpi.com/2071-1050/15/8/6657>

Torres Ner, N., et al. (2023). Measuring resilience in flood-prone communities. *Sustainability*, 15(8), 6953. <https://www.mdpi.com/2071-1050/15/8/6953>

Dulawan, J. M. T., et al. (2024). Household resilience in flood-prone areas. *Water*, 16(6), 799. <https://www.mdpi.com/2073-4441/16/6/799>

Kurata, Y. B., et al. (2023). Disaster preparedness and resilience. *Sustainability*, 15(8), 6657. <https://www.mdpi.com/2071-1050/15/8/6657>

Calapini, W. D., et al. (2025). Flood hazard modeling and resilience planning. *Geomatics*, 5(4), 64. <https://www.mdpi.com/2673-7418/5/4/64>

Singh, S. R., et al. (2024). Risk assessment and resilience strategies. *GeoHazards*, 5(4), 1040–1073. <https://www.mdpi.com/2624-795X/5/4/50>

Lauer, H., et al. (2024). Urban resilience and managed retreat. *Natural Hazards and Earth System Sciences*. <https://nhess.copernicus.org/articles/24/2243/2024/>

Dulawan, J. M. T., et al. (2024). Flood vulnerability in Metro Manila. *Water*, 16(6), 799. <https://www.mdpi.com/2073-4441/16/6/799>

Singh, S. R., et al. (2024). Flood risk mapping Philippines. *GeoHazards*, 5(4), 1040–1073. <https://www.mdpi.com/2624-795X/5/4/50>

Calapini, W. D., et al. (2025). Flood hazard mapping Cagayan River Basin. *Geomatics*, 5(4), 64. <https://www.mdpi.com/2673-7418/5/4/64>

Flood risk awareness in Marikina (2024). *International Journal of Disaster Risk Reduction*. <https://www.sciencedirect.com/science/article/pii/S2212420924005818>

Kurata, Y. B., et al. (2023). Flood preparedness Philippines. *Sustainability*, 15(8), 6657. <https://www.mdpi.com/2071-1050/15/8/6657>

Social vulnerability framework Metro Manila (2024). *International Journal of Disaster Risk Reduction*. <https://www.sciencedirect.com/science/article/pii/S2212420924005405>

Bagyong Kristine study (2025). Lived experiences of flood victims in Bicol. *Natural Hazards Research*. <https://www.sciencedirect.com/science/article/pii/S2666592125000174>

Torres Ner, N., et al. (2023). Community-level resilience narratives. *Sustainability*, 15(8), 6953. <https://www.mdpi.com/2071-1050/15/8/6953>

Dulawan, J. M. T., et al. (2024). Household perspectives in flood zones. *Water*, 16(6), 799. <https://www.mdpi.com/2073-4441/16/6/799>

Kurata, Y. B., et al. (2023). Behavioral insights on disaster response. *Sustainability*, 15(8), 6657. <https://www.mdpi.com/2071-1050/15/8/6657>

Cruz, N. A. H., & Colting-Pulumbarit, C. (2023). Post-disaster recovery narratives. *Community Development Journal*. <https://www.cddjournal.org/article/view/vol05-iss2-1>

Lauer, H., et al. (2024). Community relocation experiences. *Natural Hazards and Earth System Sciences*. <https://nhess.copernicus.org/articles/24/2243/2024/>

Flood evacuation behavior Philippines (2024). *International Journal of Disaster Risk Reduction*. <https://www.sciencedirect.com/science/article/pii/S2212420924002140>

Flood awareness Marikina (2024). *International Journal of Disaster Risk Reduction*. <https://www.sciencedirect.com/science/article/pii/S2212420924005818>

Flood vulnerability assessment Metro Manila (2024). *International Journal of Disaster Risk Reduction*. <https://www.sciencedirect.com/science/article/pii/S2212420924005405>

Singh, S. R., et al. (2024). Community-level flood risk experiences. *GeoHazards*, 5(4), 1040–1073. <https://www.mdpi.com/2624-795X/5/4/50>